Hints: Methods of proof – 6. Contradiction and proofs of impossibility

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Write each statement with all quantifiers explicit, then push the negation through, swapping $\forall \leftrightarrow \exists$ and negating the innermost statement. For example: "there is a continuous f on $[0,1]$ such that for every $x_0 \in [0,1]$ there is x with $f(x) > f(x_0)$".

Hint 2. The others: $\exists \varepsilon > 0\ \forall \delta > 0\ \exists x, y$ with $|x - y| < \delta$ and $|f(x) - f(y)| \geqslant \varepsilon$ (for the continuity-type statement); and "(a_n) diverges": $\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N\ \exists n \geqslant N$ with $|a_n - L| \geqslant \varepsilon$.

Hint 3. $\sqrt{2}$, contrapositive form: "for all integers p, $q \neq 0$ we have $p^2 \neq 2q^2$", because the exponent of 2 in p^2 is even and in $2q^2$ is odd. Descent form: if $p^2 = 2q^2$ with $p, q > 0$, then $p = 2p'$ and $q^2 = 2p'^2$ with $0 < q < p$, a smaller solution; there is no infinite descent in $\mathbb{N}$.

Hint 4. The mathematics in both is the same fact about the prime 2 (unique factorisation / parity); the contrapositive version states it directly as a valuation count, and the descent version hides it in "p even $\Rightarrow$ smaller solution".

Problem 2

Hint 1. Let G be the set of regular elements, $|G| = 5$. Check that a product of regular elements is regular and that for $a \in G$ the map $x \mapsto ax$ is injective on the whole ring. What structure does G then have?

Hint 2. G is a finite cancellative semigroup, hence a group. Its identity e is an idempotent regular element; from $e(ex - x) = 0$ and $(xe - x)e = 0$ get $ex = xe = x$, so $e = 1$ is the identity of the ring and all regular elements are units. Now $-1 \in G$ and $(-1)^2 = 1$ in a group of order 5, so $-1 = 1$: characteristic 2.

Hint 3. G is cyclic, generated by $g \neq 1$ with $g^5 = 1$. The subring $\mathbb{F}_2[g]$ is a quotient of $\mathbb{F}_2[x]/(x^5 - 1) \cong \mathbb{F}_2 \times \mathbb{F}_{16}$ (the factor $x^4 + x^3 + x^2 + x + 1$ is irreducible over $\mathbb{F}_2$). Since $g \neq 1$, the quotient is $\mathbb{F}_{16}$ or the whole product; either way it has 15 units, which are units, hence regular, in the ring. Contradiction.

Problem 3

Hint 1. First the limit: $s_n = \sum_{k \leqslant n} 1/k^2$ is bounded, and a rational function bounded on infinitely many integers has a finite limit at ∞. So $D(n) = R(n) - s_n$ converges as $n \to \infty$, and since $D(n) = 0$ infinitely often, $D(n) \to 0$.

Hint 2. Put $Q(x) = R(x) - R(x-1) - 1/x^2$, a rational function, so that $D(k) - D(k-1) = Q(k)$ and $D(n) = -\sum_{k>n} Q(k)$. If $Q \not\equiv 0$, then $Q(k)$ has a constant sign for large k, so $D(n) \neq 0$ for all large n: contradiction. Hence $R(x) - R(x-1) = 1/x^2$ identically.

Hint 3. R is not a polynomial (the difference of a polynomial is a polynomial). Let p be a pole of R with the largest real part: then $R(x-1)$ has a pole at $p+1$ where $R(x)$ is regular, so the left side has a pole at $p+1$, i.e. $p + 1 = 0$. Similarly a pole q with the smallest real part gives $q = 0$. But $\operatorname{Re} q \leqslant \operatorname{Re} p = -1$: contradiction.

Problem 4

Hint 1. Look only at the boundary circle of the unit disc. What part of it can one disc of radius $r < 1$ cover?

Hint 2. The intersection of the unit circle with a disc of radius $r < 1$ is a (possibly empty) arc, any two points of which are at distance $\leqslant 2r < 2$. So it contains no pair of antipodal points, hence it is a closed arc of angular length $< \pi$.

Hint 3. Two closed arcs of length $< \pi$ each have total length $< 2\pi$ and cannot cover the circle.

Problem 5

Hint 1. $\log_2 3 = p/q$ with $p, q > 0$ means $2^p = 3^q$: even = odd. $\sqrt{2} + \sqrt{3} = r \in \mathbb{Q}$ gives $r^2 = 5 + 2\sqrt{6}$, so $\sqrt{6} \in \mathbb{Q}$, impossible since 2 divides 6 exactly once (or: $\sqrt{2} + \sqrt{3}$ is a root of $x^4 - 10x^2 + 1$, which has no rational roots ± 1).

Hint 2. For $e = p/q$: $N = q!\left(e - \sum_{k \leqslant q} 1/k!\right)$ is an integer because $q!e = (q-1)!\,p$, and $0 < N = \sum_{k>q} \frac{q!}{k!} < \frac{1}{q+1} + \frac{1}{(q+1)^2} + \cdots = \frac{1}{q} \leqslant 1$.

Hint 3. Classification: integrality of N uses that e is a ratio p/q (the denominator q is cleared by $q!$), while the estimate $0 < N < 1$ is pure analysis; for $\log_2 3$ the ratio is used to produce the integer equation $2^p = 3^q$. In the $\sqrt{2} + \sqrt{3}$ argument the manipulation $r \mapsto (r^2 - 5)/2$ uses only field operations (it works for r in any field containing $\mathbb{Q}$, i.e. it shows $\mathbb{Q}(\sqrt{2} + \sqrt{3}) \ni \sqrt{6}$); only the final step $\sqrt{6} \notin \mathbb{Q}$, or the rational root test, uses the ratio of integers, and the degree argument ($[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = 4$) shows even more: it is not of degree $\leqslant 3$.